

DAY-2

Diagnosing hardware failures theory

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Here are five common desktop PC hardware issues with their typical symptoms:

1. Power Supply Unit (PSU) Failure

Typical symptom: PC won't turn on at all, or it suddenly shuts down/reboots under load with no warning (often no BSOD).

2. RAM (Memory) Failure

Typical symptom: System fails to POST (beep codes on startup), frequent freezes, or repeated Blue Screen of Death (BSOD) errors, often with memory-related codes.

3. Hard Drive / SSD Failure

Typical symptom: Very slow boot, clicking/grinding noises (HDD), files becoming corrupted or inaccessible, or the OS not being detected at all.

4. GPU (Graphics Card) or Display Path Failure

Typical symptom: No display on monitor despite PC seeming to power on (fans spin, lights on), or visual artifacts/screen flickering when the system is under graphics load.

5. Overheating (CPU/GPU or general thermal issue)

Typical symptom: System runs fine for a while, then throttles performance, fans become very loud, and the PC randomly shuts down or restarts, especially during gaming or heavy workloads.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

- **Check outlet** – Plug a lamp/phone charger into the same socket; if it doesn't work, the outlet is bad.[forums](#).
- **Inspect charger & cables** – Look for cuts, fraying, burn marks, bent pins, or loose connections; replace if damaged.
- **Test charger voltage (if you have a multimeter)** – Measure DC output at the plug; $\sim 0\text{ V}$ or wildly fluctuating = bad charger.[forums](#).
- **Swap with a known-good compatible charger** – If the laptop charges, original charger is faulty; if not, problem is in the laptop.

- *Check laptop DC jack* – Ensure plug fits snugly, no debris/bent pins; if charging only works at certain angles, the jack is likely damaged.
- *Hard reset / reseal battery* – Power off, unplug, remove battery (if possible), hold power 30 s, reinstall battery, reconnect charger.
- *Decide next step* – Bad outlet/cable → fix power source; bad charger → replace; bad jack/internal → repair/replace hardware.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

How to measure it:

1. Set your multimeter to **DC voltage** mode, ideally the 20V range (not 200V or 2V — 20V gives good resolution for a 5V signal).
2. Plug the charger into a wall outlet, but don't plug the USB cable into a phone.
3. Touch the **red (positive) probe** to the USB port's power pin (the outer left pin, pin 1, on a USB-A port) and the **black (negative) probe** to the ground pin (outer right pin, pin 4).
4. If you're using a USB cable with exposed wires or a USB breakout/test connector, it's easier — red probe to the red wire, black probe to the black wire.
5. Read the display.

What to expect:

- *A healthy USB charger should read close to 5.0V, and anywhere from 4.75V–5.25V is considered within spec (USB allows about $\pm 5\%$ tolerance).*
- *Under a real load (charging a phone), voltage typically sags slightly — 4.8V–5.0V is normal.*
- *No reading (0V) usually means a dead charger, bad outlet, or you've got the probes on the wrong pins.*
- *A reading noticeably above 5.25V (some fast-chargers can jump to 9V or 12V under certain negotiation protocols like Quick Charge/PD) is normal only if the charger is in a fast-charge handshake — for a baseline "standard 5V" test, use a charger without fast charging, or measure with no data lines connected.*

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Steps:

1. *Power off and unplug the PC.*
2. *Disconnect the 24-pin ATX connector from the motherboard.*
3. *Bridge the green wire (PS_ON) to a black wire with a paperclip to force the PSU on.*
4. *Plug the PSU back in — fan should spin.*
5. *Set multimeter to DC voltage, 20V range.*

6. Probe red lead on a yellow wire, black lead on a black wire. Read the value.

Expected: ~11.4V-12.6V is in spec (nominal 12V $\pm$ 5%).

I don't have a real PSU to test, so I can't give you an actual measured number — you'd need to read it off your multimeter.

If below 11.5V: likely a failing/aging power supply (weak capacitors), an overloaded PSU that can't sustain the rail under load, or a bad/corroded connector. Since CPU and GPU draw heavily on 12V, a sagging rail is a common cause of random shutdowns under load.